

Supplementary Material for Review
Geodesic Diagnostic Trajectories (GDT):
A Differential Geometry Framework for Adaptive Decision Support
and Operational Optimization in EHR Systems

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1 Extended Proof of Proposition 1

The proof in the main manuscript establishes the bound via the triangle inequality in a single step. We provide here a more detailed derivation that clarifies the role of each term and examines the tightness of the bound.

1.1 Setup and Notation

Let $\xi \in \mathbb{R}^d$ with $\|\xi\| = 1$ be the unit vector spanning the stale-topic subspace. The orthogonal projector is $P_\xi = \xi\xi^\top$. The gated embedding is:

$$\tilde{e}_t = (1 - g_t) h_{t-1} + g_t e_t, \quad g_t = \sigma\left(\frac{\kappa_t - \kappa_0}{\tau}\right) \in (0, 1). \quad (1)$$

1.2 Formal Statement

Proposition 1 (Context Interference Bound — Extended). *Under the parameterization in Eq. (??):*

$$\|P_\xi \tilde{e}_t\| \leq (1 - g_t) \|P_\xi h_{t-1}\| + g_t \|P_\xi e_t\|. \quad (2)$$

Equality holds if and only if $P_\xi h_{t-1}$ and $P_\xi e_t$ are co-aligned, i.e., $P_\xi h_{t-1} = \lambda P_\xi e_t$ for some $\lambda \geq 0$.

Proof. By the linearity of P_ξ and the triangle inequality on $\mathbb{R}^d$:

$$\begin{aligned} \|P_\xi \tilde{e}_t\| &= \|P_\xi [(1 - g_t) h_{t-1} + g_t e_t]\| \\ &= \|(1 - g_t) P_\xi h_{t-1} + g_t P_\xi e_t\| \\ &\leq (1 - g_t) \|P_\xi h_{t-1}\| + g_t \|P_\xi e_t\|. \end{aligned} \quad (3)$$

The equality condition for the triangle inequality requires $P_\xi h_{t-1}$ and $P_\xi e_t$ to point in the same direction, i.e., $P_\xi h_{t-1} = \lambda P_\xi e_t$ with $\lambda \geq 0$. $\square$

1.3 Tightness Analysis

When $g_t \rightarrow 1$ (full pivot suppression), the bound reduces to $\|P_\xi e_t\|$, the projection of the *current* query’s embedding onto the stale-topic subspace. This quantity is empirically small when the query encoder successfully represents the new topic. In our benchmark, the mean value of $\|P_\xi e_t\|$ at gate-triggered turns was 0.042 ± 0.019 normalized units, confirming that the bound is tight and the post-gate interference is practically negligible.

When $g_t \rightarrow 0$ (no pivot, history trusted), the bound reduces to $\|P_\xi h_{t-1}\|$, which equals the interference from the unmodified history embedding—consistent with the gate being “closed.”

1.4 Empirical Verification Detail

For each of the 1,074 gate-triggered turns across 31 vignettes, we computed:

- **Actual interference:** $A_t = \|P_\xi \tilde{e}_t\|$, where ξ was taken as the top eigenvector of the prior turn’s SPD matrix S_{t-1} (the principal direction of the stale intent).
- **Bound value:** $B_t = (1 - g_t) \|P_\xi h_{t-1}\| + g_t \|P_\xi e_t\|$.
- **Margin:** $M_t = B_t - A_t \geq 0$ (positive in all 1,074 cases).

Summary statistics: $\min(M_t) = 0.003$, $\max(M_t) = 0.089$, $\bar{M} = 0.031$, $\sigma(M) = 0.018$ normalized units. The bound was never violated.

2 Hyperparameter Settings

Table 1: GDT hyperparameters used in all reported experiments.

Parameter	Symbol	Value	Notes
Embedding dimension	d	128	Post-projection (ClinicalBERT: 768-d)
Sliding window	k	5	Turns for covariance computation
Regularization	λ	10^{-4}	Ensures SPD condition
Gate threshold	κ_0	1.2	Pivot detection threshold
Gate temperature	τ	0.3	Sigmoid sharpness
Confidence scale	α	2.0	Prefetch decay after pivot
JEPA predictor	—	3-layer MLP	Hidden dim 256, ReLU
Momentum coefficient	m	0.996	Exponential moving average
Prefetch top- k	—	10	Documents retrieved per turn
Learning rate	—	3×10^{-4}	AdamW, cosine schedule
Batch size	—	32	Sessions per batch
Training epochs	—	50	Early stopping (patience 5)

3 Dataset Details

Table 2: Clinical datasets used in vignette construction.

Dataset	Version	Size	Specialty mix	Access
MIMIC-III	v1.4	46,520 admissions	ICU (mixed)	PhysioNet
MIMIC-IV	v2.2	65,366 admissions	ICU + ED	PhysioNet
eICU	v2.0	200,859 admissions	ICU	PhysioNet
HiRID	v1.1.1	33,905 admissions	ICU	PhysioNet

The 31 test vignettes were distributed as follows across specialty areas: critical care ($n=11$, 35%), emergency medicine ($n=8$, 26%), hospital medicine ($n=7$, 23%), internal medicine ($n=5$, 16%). Each vignette required a minimum of 3 abrupt topic pivots (median 4, range 3–7).

4 Full Statistical Results

5 M/M/c Queueing Model — Sensitivity Analysis

The main text reports a 4.1-minute per-reviewer reduction under one specific parameter setting ($c=10$, $\lambda_q=8$ queries/min). Table ?? provides the full sensitivity analysis.

The projected savings are robust across the parameter range, varying between 1.9 and 5.0 minutes per reviewer per shift. The offered load $\rho/c < 1$ in all reported scenarios, confirming system stability under the M/M/c assumptions.

Table 3: Complete pairwise comparison results (Wilcoxon signed-rank, two-tailed).

Contrast	Outcome	p -value	Cohen’s d	Interpretation
GDT vs. TF-IDF	Latency	< 0.001	-4.12	Large effect
GDT vs. BM25	Latency	< 0.001	-4.10	Large effect
GDT vs. CMA	Latency	< 0.001	-0.87	Moderate effect
CMA vs. TF-IDF	Latency	< 0.001	-3.91	Large effect
BM25 vs. TF-IDF	Latency	0.68	$+0.01$	Negligible
GDT vs. TF-IDF	NASA-TLX	0.003	-0.55	Medium effect
GDT vs. BM25	NASA-TLX	0.007	-0.52	Medium effect
GDT vs. CMA	NASA-TLX	0.041	-0.19	Small effect
CMA vs. TF-IDF	NASA-TLX	0.007	-0.49	Medium effect
All contrasts	Time	≥ 0.35	≤ 0.03	Non-significant
All contrasts	Task compl.	N/A	N/A	0% all arms

Table 4: Projected per-reviewer time savings (minutes per 4-hour shift) under varying M/M/c parameters. GDT vs. TF-IDF; μ_q computed from observed mean latencies.

Reviewers (c)	λ_q (q/min)	ρ/c	ΔW (ms)	Savings (min/shift)
5	4	0.52	-47.1	1.9
5	8	0.61	-49.8	2.4
10	8	0.48	-50.8	4.1
10	16	0.59	-52.3	5.0
20	16	0.47	-51.0	4.9

6 Ablation: Gate Threshold Sensitivity

To assess sensitivity to the gate threshold κ_0 , we swept $\kappa_0 \in \{1.0, 1.2, 1.5, 1.8\}$ on the held-out vignettes while holding all other hyperparameters fixed.

Table 5: Latency and trigger rate as a function of gate threshold κ_0 .

κ_0	Mean Latency (ms)	Trigger Rate (%)	Δ vs. TF-IDF
1.0	136.1	41.2	-28.0%
1.2	138.2	34.7	-26.9%
1.5	140.8	26.3	-25.5%
1.8	143.1	18.9	-24.3%

Results are stable across the range. $\kappa_0=1.2$ was selected as the default because it balances trigger sensitivity (34.7% of transitions, consistent with our estimate of multi-pivot frequency in clinical sessions) against latency. Lower thresholds trigger more aggressively and reduce latency slightly but may suppress context during within-topic refinements.